

Large language models' representations of monetary value in dictator games

Qianze Zhang¹, Pim Heijnen¹, and Zsombor Méder¹

¹Department of Economics, Econometrics and Finance, University of Groningen

Abstract

Large language models (LLMs) show promise for and are increasingly deployed in decision-making contexts, yet their processing of monetary incentives remains poorly understood. We study LLM responses in two-alternative dictator game choice tasks, and model choices using a random linear utility model within a psychophysical framework. We study LLMs' sensitivity to monetary differences at different scales by systematically varying the budget and payoff differences. Our results reveal a surprising pattern: while models of intermediate size exhibit logarithmic compression consistent with Weber's Law, this regularity is not universal. Specifically, models with extensive reasoning capabilities rely on absolute value comparisons, suggesting that psychophysical laws in LLMs depend on model capacity and architecture.

Keywords: large language models, decision-making, value representation.

Introduction

Following the rapid development of reasoning ability in large language models (LLMs) (Kojima et al., 2022), social scientists have begun adapting methods from economics and psychology in order to better understand these systems (Chen et al., 2023; Guo, 2023; Horton, 2023; Manning et al., 2024). One prominent line of work, building upon recent advances in LLM-based agents and multi-agent decision-making frameworks (Bouneffouf & Feraud, 2025; Huang et al., 2025; N. Li et al., 2024; Yao et al., 2022), places large language models in game-theoretic and multi-agent environments to observe their behavioral patterns (Akata et al., 2025; Fan et al., 2024). Nevertheless, the design of such interactions typically relies on the textual specification of incentives expressed in monetary terms. This raises a fundamental question: when LLMs engage in tasks involving allocations or investment decisions, how do they internally represent monetary value?

Understanding how LLMs internally represent money is not only a theoretical concern, but also has direct implications to their real-world applicability. LLMs are increasingly

Qianze Zhang  <https://orcid.org/0009-0006-7099-2393>

Correspondence concerning this article should be addressed to Qianze Zhang, Department of Economics, Econometrics and Finance, University of Groningen. Email: qianze.zhang@rug.nl

used as decision-support systems, autonomous agents, and as predictors or simulators of human decision-making in domains involving pricing, negotiation, resource allocation, and financial planning (Bianchi et al., 2024; Fedyk et al., 2024; Jia et al., 2024; Lai et al., 2025; Wachter et al., 2025). In such contexts, coherent decision-making critically depends on how monetary magnitudes are internally encoded and compared. If an agent systematically distorts monetary differences – for example, by overreacting to small stakes or underreacting to large ones – its decisions may appear inconsistent, biased, or misaligned with human expectations, even when its reasoning is otherwise sophisticated. Studying monetary representation therefore provides a principled entry point into the internal decision-making mechanisms of LLMs. Such distortions motivate a psychophysical perspective, in which internal representations are inferred from systematic patterns in choice behavior. This motivates a psychophysical approach: as in the study of perception, internal representations can be inferred from systematic relationships between stimuli and behavior (Gescheider, 2013). Here, varying monetary magnitudes and observing choices allows one to probe how monetary value is internally encoded.

In psychophysics, there is a long tradition of studying how humans represent objective quantities through subjective scaling. One robust finding is that human perception often follows logarithmic compression, i.e., equal proportional changes in objective magnitude are perceived as equal subjective differences, as formalized by Weber’s law (Fechner, 1860; Weber, 1834). Since LLMs are trained on large amounts of human-generated data (Radford et al., 2019) and therefore implicitly learn from human behavioral patterns (Ross et al., 2024), it is natural to ask whether similar perceptual regularities emerge in these models. At the same time, such internal representations are not deliberately imposed by model architecture or training objectives, and are difficult to inspect directly (Bommasani, Hudson, et al., 2022; Olah et al., 2018). Recent evidence suggests that LLMs exhibit Weberian scaling in the context of time perception (L. Li et al., 2025). Nevertheless, whether this observed regularity also holds for monetary perception is not known. Given that money is central to almost all economic interactions and games, answering this question is of fundamental importance.¹

We investigate this question using a psychophysically motivated experimental design using a dictator game setting. Our research centers on two questions. First, how do LLMs scale monetary differences across varying levels of absolute magnitude? Second, do model characteristics and sampling temperature affect the underlying representation of monetary value, or do they primarily influence the stochasticity of observed choice behavior? To address these questions, we employ a linear random utility framework from discrete choice theory (Anderson et al., 1992).

The paper is structured as follows: In Section , we present our theoretical framework that combines a linear random utility model with a subjective monetary representation using a psychometric choice function. Section describes the experimental design, which consists of a Dictator Game using a two-alternative forced-choice task. In Section , we report our parameter estimation and robustness analyses. Section summarizes and discusses our

¹Related perspectives from mechanistic interpretability emphasize an analogy to biological systems, arguing that understanding intelligent behavior requires studying internal mechanisms rather than input–output behavior alone; see the Transformer Circuits project by Anthropic (<https://transformer-circuits.pub/2025/attribution-graphs/biology.html>).

findings, suggesting further avenues.

Theoretical Framework

Subjective Value Representation

We consider a binary choice task concerning a total available budget of m . The decision maker (i.e., the 'dictator') chooses between two allocations $C_i = (a_i, b_i)$, $i \in \{1, 2\}$, satisfying $a_i + b_i = m$. Here, a_i and b_i denote the amounts received by the decision maker and the recipient, respectively. Allocation C_1 is an equal split, with $a_1 = b_1 = \frac{m}{2}$, whereas C_2 is unequal, with $a_2 \neq b_2$.

Let $d = a_2 - a_1$ denote the gain, defined as the change in the dictator's payoff relative to the equal split. A positive value of d thus indicates that option C_2 provides a higher payoff to the dictator, i.e., that they can appropriate a larger share of the available budget.

Following the psychophysical approach to magnitude perception, we assume that decisions are based on a subjective value representation that may depend on the properties of the choice set—here, budget size and the incentive to deviate from the equal split. We define the subjective value signal as:

$$V(d, m) = \frac{d}{m^\gamma}, \quad \text{with } \gamma \geq 0.$$

Here, γ is a scaling exponent that governs how monetary differences are compressed as a function of the overall magnitude. This formulation encompasses two edge cases. First, when $\gamma = 0$, the subjective value signal becomes simply $V = d$, corresponding to absolute scaling in which monetary differences are perceived identically, independently of choice context. Alternatively, under $\gamma = 1$, subjective value becomes $V = \frac{d}{m}$, corresponding to Weberian scaling, in which perception depends on relative rather than absolute differences and implies logarithmic compression. We treat the exponent γ as a structural parameter, and will estimate it from observed choice data in Section .

Psychometric Choice Function

We model the probability of choosing the equal allocation C_1 as a logistic function of the subjective value signal (Gescheider, 2013):

$$\Pr(C_1 \mid d, m) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 V(d, m))}}.$$

The intercept parameter β_0 captures baseline bias towards allocation C_1 when under no monetary gains $d = 0$. The slope parameter β_1 describes sensitivity to subjective monetary differences, i.e., it indicates how fast probability changes around indifference. A larger absolute value of β_1 corresponds to sharper discrimination.

The point of subjective equality (PSE), defined as the subjective value at which the choice probabilities are equal, $\Pr(C_1) = 0.5$, is given by:

$$PSE_V = -\frac{\beta_0}{\beta_1}.$$

This quantity represents the subjective indifference point between the two allocation options.

Mapping this indifference condition back to objective monetary space yields the corresponding threshold:

$$d^*(m) = -\frac{\beta_0}{\beta_1} m^\gamma.$$

At the level of internal representation, $\gamma > 0$ indicates context-dependent value scaling: monetary differences are normalized by the overall budget magnitude before entering the decision process. Importantly, this scaling is not imposed by model architecture or training objectives, but instead emerges from the model's learned representations. As a result, the same objective monetary difference can be encoded differently depending on its contextual magnitude.

Linear Random Utility Model

While the psychometric choice function provides a convenient reduced-form description of choice behavior, it remains agnostic about the underlying decision process generating these probabilities. To connect the psychophysical specification to a standard decision-theoretic framework, we derive it from a linear random utility model (LRUM) from discrete choice theory (Anderson et al., 1992). This micro-foundation allows us to interpret the estimated coefficients in terms of latent preferences and stochasticity, and to clarify the role of temperature as a source of noise rather than value distortion.

The utility of option is given by

$$U_j = \bar{U}_j + \varepsilon_j, j \in \{1, 2\}.$$

where $\bar{U}_j$ is the deterministic component of choosing C_j and ε_j is an i.i.d. stochastic term capturing generation noise. Assuming ε_j follows a Type-I extreme value (Gumbel) distribution with scale parameter μ , the probability of choosing the fair option is

$$\Pr(C_1) = \frac{e^{\bar{U}_1/\mu}}{e^{\bar{U}_1/\mu} + e^{\bar{U}_2/\mu}} = \frac{1}{1 + e^{(\bar{U}_2 - \bar{U}_1)/\mu}} = \frac{1}{1 + e^{\bar{U}_d/\mu}}.$$

where $\bar{U}_d = \bar{U}_2 - \bar{U}_1$. This assumption yields the familiar logit choice rule, which is formally equivalent to the psychometric function used in the previous subsection.

We parameterize the deterministic utility difference as:

$$\bar{U}_d = \frac{d}{m^\gamma} - g.$$

For parsimony, we normalize the marginal utility of money to 1, absorbing scale differences into the noise parameter.² The utility cost associated with deviating from the equal allocation choice is captured by g .

Substituting yields

²In the present specification, the marginal utility of monetary differences is normalized without loss of generality. In binary choice models, preference strength and stochasticity enter the choice probability only through their ratio, so an explicit scale parameter is not separately identifiable. Accordingly, we absorb any constant marginal utility factor into the noise scale and focus on the resulting sensitivity parameters.

$$\Pr(C_1) = \frac{1}{1 + e^{\frac{(d/m^\gamma) - g}{\mu}}} = \Lambda\left(\frac{g}{\mu} - \frac{1}{\mu} \frac{d}{m^\gamma}\right).$$

Comparing the two representations implies

$$\begin{aligned}\beta_0 &= \frac{g}{\mu}, \\ \beta_1 &= -\frac{1}{\mu}.\end{aligned}$$

Thus, the estimated regression coefficients correspond to preference-to-noise ratios: both baseline bias and monetary sensitivity reflect the interaction between underlying preferences and the scale of stochasticity.

In large language models, the generation temperature T directly controls output randomness. We interpret temperature as proportional to the noise scale parameter:

$$\mu \propto T.$$

Consequently,

$$\beta_1(T) \propto -\frac{1}{T}.$$

As $T \rightarrow 0$, noise vanishes and choices become deterministic, yielding step-function psychometric curves. As T increases, sensitivity decreases and choices approach random responses. Importantly, temperature affects stochasticity but not the scaling exponent γ , which governs subjective value representation.

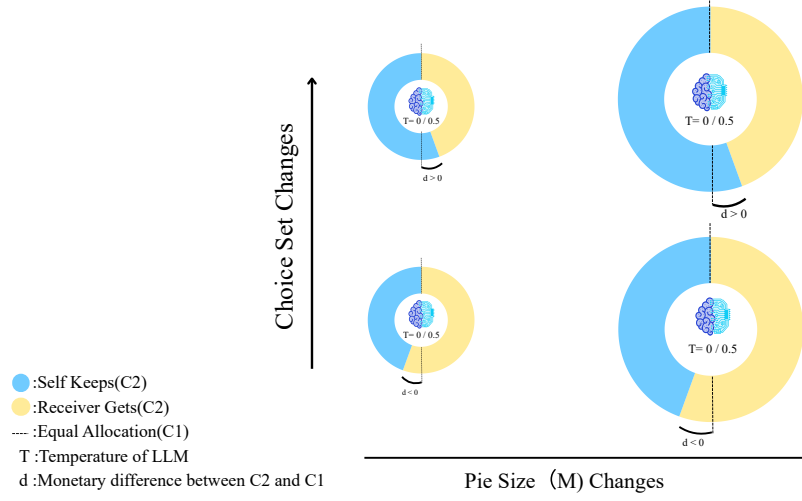
This mapping clarifies that the psychometric parameters β_0 and β_1 jointly reflect underlying preferences and noise, while the scaling exponent γ governs how monetary differences are internally represented before stochastic choice is applied.

Method

Experimental Design

To systematically investigate the behavior of large language models (LLMs) in the dictator game, we adopt an experimental design that varies the total available budget, following the approach proposed in (Schulz et al., 2014). Varying the budget allows us to examine how allocation decisions respond to changes in the order of magnitude of monetary resources while holding the relative structure of the choice constant. This manipulation is essential for distinguishing absolute from relative value sensitivity and for testing whether monetary perception follows a Weberian scaling pattern.

Our experimental design is also based on the classical two-alternative forced-choice (2AFC) paradigm, which is widely used in psychophysics to study perceptual discrimination and value comparison (Liang & Acuna, 2020). In a 2AFC task, participants are repeatedly presented with two alternatives and are required to choose exactly one option on each trial, yielding a binary choice outcome. This paradigm enables researchers to infer latent perceptual or value representations from observed choice behavior under controlled stimulus variation.

**Figure 1**

Schematic of choice sets in the dictator game. The vertical axis shows variations in monetary difference d (choice set changes) at a fixed pie size. The horizontal axis shows variations in pie size m at a fixed relative difference. Icons represent the LLM as the dictator, with temperature T influencing choice stochasticity.

In each trial, the LLM is presented with two allocation options: an equal allocation (C_1) and an alternative allocation (C_2) that either favors or disfavors the dictator by a monetary difference d (with positive d indicating higher payoff for the dictator under C_2). Figure 1 presents the general idea of the allocation design. By systematically varying d across trials while holding the framing, prompt structure, and number of options fixed, we can isolate how choice behavior responds to changes in monetary incentives. For each trial, the model must select exactly one of the two available options. A representative sample of the experimental stimuli and the full prompt template are provided in Appendix A.

Procedure

The experiments were conducted using four large language models: deepseek-r1:70B³, llama3.1:70B⁴, gpt-oss:20b⁵, and llama3.1:8b⁶. We constructed 66 distinct choice sets for the dictator game, each consisting of an equal allocation (C_1) and an unequal allocation (C_2) differing by a monetary amount. To avoid systematic labeling effects, the allocation options in each trial were randomly assigned to option labels A and B in the prompt. Each choice set was repeated 10 times using an identical prompt, yielding repeated observations under controlled conditions. Each model was evaluated under two values of the temperature parameter, $T = 0$ and $T = 0.5$. All models were run locally.

Each trial is conducted as a one-shot decision problem even all trials under one

³<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-70B>.

⁴<https://huggingface.co/meta-llama/Llama-3.1-70B>.

⁵<https://huggingface.co/openai/gpt-oss-20b>.

⁶<https://huggingface.co/meta-llama/Llama-3.1-8B>.

temperature condition were run at once. LLMs are explicitly instructed to ignore any previous interactions and to base their choice solely on the information presented in the current trial.

Model outputs were restricted to a binary response format, allowing only option A or option B. After we collected the raw data, we excluded responses that could not be unambiguously classified as either option A or option B from the analysis. Each observation corresponds to a single choice made by a model in one trial.

Results

Weberian scaling of monetary differences

We first estimate the psychophysical scaling exponent γ governing how monetary differences are represented. For each model and temperature condition, we perform a profile likelihood search over $\gamma \in [0, 1.5]$, selecting the value that minimizes the Akaike Information Criterion (AIC). Table 1 summarizes the resulting estimates.

Table 1

AIC-based grid search for the scaling exponent γ by model and temperature. $\bar{y}$ denotes the mean probability of choosing the equal allocation C_1 .

Model	$\bar{y}$	$\hat{\gamma}$	AIC
deepseek-r1:70b, T=0.0	0.873	1.05	499.8
deepseek-r1:70b, T=0.5	0.852	1.00	535.2
gpt-oss:20b, T=0.0	0.409	1.05	335.0
gpt-oss:20b, T=0.5	0.445	0.95	523.1
llama3.1:70b, T=0.0	0.364	1.00	4.0
llama3.1:70b, T=0.5	0.394	0.95	510.7
llama3.1:8b, T=0.0	0.476	0.95	914.9
llama3.1:8b, T=0.5	0.461	0.80	901.5

Across all models and temperatures, the best-fitting scaling exponent clusters closely around $\gamma \approx 1$. This pattern holds consistently for all LLMs, although with slightly larger deviations at higher temperature. The concentration of best-fit values near $\gamma = 1$ indicates that LLMs scale monetary differences approximately in proportion to the relative difference d/m , consistent with Weberian (logarithmic) scaling rather than absolute value comparison.

Then we estimate the parameter β_1 , using the best-fitting γ for each condition of each model. Table 2 reports the estimated coefficients and standard errors. The estimated β_1 values are generally negative, reflecting the task structure: positive monetary differences favor the alternative allocation C_2 , thereby reducing the probability of choosing the fair option C_1 .

The magnitude of β_1 varies substantially across models. Larger models such as llama3.1:70b under Temperature 0 exhibit extremely steep slopes at low temperature, indicating near-deterministic choice behavior. In contrast, llama3.1:8b displays much smaller slope magnitudes, with choice probabilities remaining close to chance levels, suggesting weak sensitivity to monetary differences under this specification.

Table 2

Estimated slope parameter β_1 for each model and temperature condition.

Larger absolute values indicate sharper choice sensitivity. β_1 is estimated using the best-fitting scaling exponent γ for each model. p -values are reported for completeness but should be interpreted cautiously given large sample sizes and instances of near-deterministic choice behavior. For llama3.1:70b at $T=0.0$, the model exhibits perfect separation. The maximum likelihood estimate for the slope diverges to infinity, rendering standard errors and p -values undefined.

Model	β_1	Std. Error	p
deepseek-r1:70b, $T=0.0$	−23.74	8.95	.008
deepseek-r1:70b, $T=0.5$	−40.95	8.79	< .001
gpt-oss:20b, $T=0.0$	−280.20	22.49	< .001
gpt-oss:20b, $T=0.5$	−165.17	11.96	< .001
llama3.1:70b, $T=0.0$	<i>N.A.</i>	<i>N.A.</i>	<i>N.A.</i>
llama3.1:70b, $T=0.5$	−120.21	8.89	< .001
llama3.1:8b, $T=0.0$	0.97	0.67	.15
llama3.1:8b, $T=0.5$	−2.51	0.71	< .001

Increasing temperature systematically reduces the absolute magnitude of β_1 for llama3.1:70b and gpt-oss:20b, resulting in greater stochasticity in choices. This pattern is consistent with the interpretation of temperature as a noise parameter that modulates choice randomness. However, this effect is less pronounced for deepseek-r1:70b and llama3.1:8b, which exhibit relatively small slope magnitudes across temperature conditions.

A small absolute value of β_1 implies weak sensitivity of choice probabilities to subjective value differences, consistent with near-random responding. For llama3.1:8b, this pattern may reflect limited capacity to extract and utilize fine-grained monetary distinctions in the task. deepseek-r1:70b, by contrast, rarely deviates from the equal allocation despite substantially longer generation times, with an average response time of 171.82 seconds—approximately 245 times longer than that of llama3.1:8b. This contrast suggests that extended computation does not necessarily translate into greater behavioral sensitivity to monetary incentives, but may instead reflect differences in internal processing or alignment objectives.

Crucially, while temperature strongly affects choice sensitivity, it does not materially alter the estimated scaling exponent γ . Across temperature conditions, the best-fitting γ values remain close to one, indicating that temperature primarily modulates decision noise rather than the underlying representation of monetary differences.

To assess baseline allocation bias independently of sensitivity, we computed the point of subjective equality (PSE), defined as the monetary difference at which the model is indifferent between the equal and unequal allocations. PSE is given by $-\beta_0/\beta_1$ from logistic fits within each pie size condition.

Across models exhibiting reliable monetary sensitivity (deepseek-r1:70b, gpt-oss:20b, and llama3.1:70b), estimated PSE values systematically increase with pie size, indicating

that larger absolute differences are required to offset the equal allocation as stakes increase. This pattern is consistent with context-dependent normalization of value differences. In contrast, PSE estimates for llama3.1:8b are highly unstable across conditions, reflecting weak and often statistically insignificant slope estimates. In such cases, PSE values should be interpreted with caution, as small changes in estimated sensitivity lead to large numerical variation in indifference thresholds. Complete PSE estimates and associated regression statistics for all models and pie sizes are reported in Appendix B.

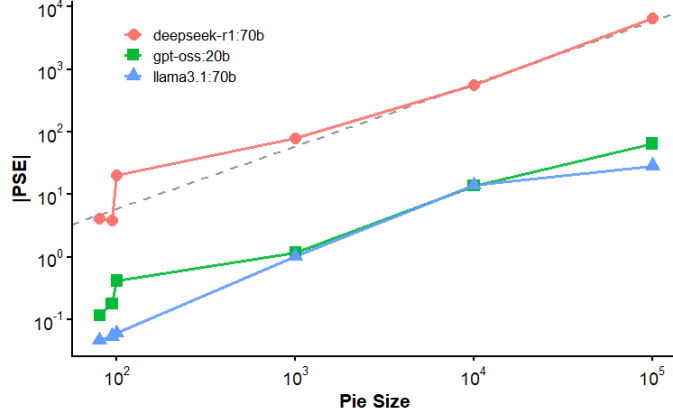


Figure 2

Scaling of Point of Subjective Equality (PSE) with Magnitude. *The absolute PSE values are plotted against Pie Size on log-log axes. The linear relationship (slope ≈ 1) confirms that the models’ perceptual thresholds scale proportionally with magnitude, consistent with Weber’s Law. llama3.1:8b is excluded due to unstable estimates.*

Since the estimation of the Point of Subjective Equality (PSE) is performed independently for each condition and does not rely on the global scaling parameter γ , it serves as a robust, model-agnostic metric for validating Weber’s Law. Theoretically, if Weber’s Law holds, the magnitude of the PSE is expected to scale linearly with the pie size (magnitude m). After excluding the llama3.1:8b model due to estimation instability, we plotted the absolute PSE values of the remaining models against pie size on logarithmic axes. As shown in Figure 2, a distinct linear relationship is observed, confirming that the models’ perceptual thresholds scale proportionally with magnitude.

The near-deterministic choice behavior observed in some large models differs sharply from typical human data, in which choices are rarely perfectly separable. This feature leads to separation issues in standard logistic estimation and motivates the robustness analyses reported in the following section.

Robustness analysis

To address potential separation in near-deterministic choice behaviors, we re-estimate the γ profile using Firth’s bias-reduced logistic regression. For each model–temperature condition and each value of γ on the same grid, we fit a Firth logit model

and record the penalized log-likelihood. The best-fitting γ is selected as the value maximizing the penalized log-likelihood.

Table 3

Comparison of γ estimates: GLM AIC profile vs. Firth log-likelihood profile. $\hat{\gamma}_{GLM}$ is selected by minimizing AIC (Table 2). $\hat{\gamma}_{Firth}$ is selected by maximizing the Firth log-likelihood over the same grid. In conditions with weak sensitivity (slope near zero) or limited response variation, the likelihood can be relatively insensitive to γ , and the grid-based maximizer may land at a boundary (e.g., $\gamma = 0$). These cases are treated as robustness diagnostics rather than primary evidence against Weberian scaling.

Model	T	$\hat{\gamma}_{GLM}$	$\hat{\gamma}_{Firth}$	LL_{max}	n
deepseek-r1:70b	0.0	1.05	0.00	-239.41	660
deepseek-r1:70b	0.5	1.00	0.00	-264.34	660
gpt-oss:20b	0.0	1.05	0.90	-178.02	657
gpt-oss:20b	0.5	0.95	0.95	-259.48	660
llama3.1:70b	0.0	1.00	0.75	-33.91	660
llama3.1:70b	0.5	0.95	0.95	-253.35	660
llama3.1:8b	0.0	0.95	0.00	-444.56	660
llama3.1:8b	0.5	0.80	0.00	-440.65	660

For gpt-oss:20b and llama3.1:70b, the standard GLM identifies $\gamma \approx 1$. The Firth-corrected estimates remain in the same range. As shown in Table C1, llama3.1:70b (at $T = 0.0$) exhibits an AIC of 4.0. This indicates near-perfect separation, where standard logistic regression fails because parameters approach infinity. gpt-oss:20b similarly displays a clear sigmoid curve. Thus, the Firth method provides a necessary correction for calibration in these cases. For llama3.1:8b, the slope β_1 is small. To prevent overfitting, the Firth method penalizes the γ estimate to 0. This result reflects low model sensitivity rather than a reliable robustness check. Intuitively, if a model is insensitive to the decision context, analyzing its internal representation is meaningless. The model likely selects options randomly, resulting in a choice probability near 0.5. deepseek-r1:70b shows a similar pattern due to extensive reasoning or a strong bias for C_1 (the safer option). This behavior implies insensitivity to numerical differences and magnitudes. Consequently, the Firth estimate reverts to 0. These cases do not prove that the models violate Weber's Law. Instead, they demonstrate that the Firth method cannot detect Weberian scaling when models lack basic sensitivity to monetary values.

Discussion

This study provides a psychophysically grounded analysis of how large language models represent monetary differences in a dictator game setting. While the main results reveal robust evidence for Weberian scaling in large models, several extreme or degenerate parameter estimates warrant careful interpretation and provide important methodological insights.

First, under Firth-corrected logistic regression, the estimated scaling exponent γ collapses to zero for deepseek-r1:70b and llama3.1:8b. These two cases, however, reflect

fundamentally different underlying behaviors. For llama3.1:8b, the distribution of choices remains close to random across monetary differences, with approximately equal probabilities of choosing the equal and unequal allocations within the same condition. In such cases, the model exhibits little systematic sensitivity to monetary variation, making it infeasible to diagnose any meaningful internal representation of monetary value. This pattern is consistent with prior findings that small-scale LLMs fail to exhibit stable psychometric structure in magnitude-based tasks, including temporal perception (L. Li et al., 2025). When no discriminative signal is present in behavior, the Firth correction necessarily yields a degenerate estimate, as there is no underlying slope to regularize.

By contrast, deepseek-r1:70b displays a strong and persistent preference for the equal allocation across most conditions. In this case, the collapse of the Firth-based γ estimate to zero reflects not stochasticity, but rather extremely low sensitivity to monetary differences due to a dominant baseline preference. In both cases, the Firth procedure does not generate additional substantive insights, as the effective slope parameter β_1 is close to zero for different reasons—random responding in one case and near-deterministic fairness preference in the other. Importantly, the Point of Subjective Equality (PSE) estimates remain more informative for deepseek-r1:70b, indicating systematic shifts in indifference points consistent with Weberian scaling in monetary representation.

Second, several results reveal extreme slope estimates in standard logistic regressions, most notably for llama3.1:70b at temperature $T = 0$. In these conditions, choices are nearly deterministic, resulting in complete or quasi-complete separation. As is well known, maximum likelihood estimation in logit models is ill-defined under perfect separation, leading to diverging coefficient estimates and inflated standard errors. This phenomenon highlights a broader methodological issue when using LLMs as experimental agents: their behavior can be substantially more deterministic than that of human participants and may lack the trial-to-trial variability implicitly assumed by standard econometric tools. Recent studies have similarly noted that LLM decision-making often exhibits reduced behavioral noise relative to humans (Chen et al., 2023; Horowitz & Plonsky, 2025). Our results therefore underscore that statistical methods developed for human data may require careful adaptation when applied to LLM-based experiments.

Despite these challenges, the investigation of monetary value representation in LLMs remains theoretically and practically valuable. Prior work has shown that LLMs exhibit Weberian compression in temporal perception (L. Li et al., 2025), suggesting that magnitude normalization emerges at the level of internal representations rather than being imposed by task structure. Our findings extend this insight to the monetary domain: for sufficiently large models, subjective sensitivity to monetary differences scales approximately proportionally with overall magnitude, consistent with Weber’s law. This convergence across domains strengthens the interpretation that LLMs encode quantitative information in a scale-invariant manner, analogous to human perceptual systems.

Taken together, while standard GLM estimates suggest broad adherence to Weber’s Law, the robustness analyses reveal a more nuanced pattern. Smaller models exhibit behavior consistent with random responding, whereas the most capable model (deepseek-r1:70b) appears to rely less on relative magnitude scaling and more on near-absolute value comparisons. This divergence suggests that Weberian scaling in LLMs may not constitute a universal property, but instead emerge within a specific range of model capacities and

training regimes.

One possible interpretation is that sensitivity to relative magnitude follows a non-monotonic pattern across model scale: insufficient capacity yields stochastic choice, intermediate capacity produces Weberian compression, and highly capable models may partially override perceptual biases through more structured reasoning. Alternatively, observed Weberian behavior may reflect properties of the training distribution rather than a fundamental constraint of model architecture. Distinguishing between these explanations remains an important direction for future work.

Finally, these results have implications for both future modeling and applied use of LLMs. From a methodological perspective, models of LLM decision-making should account not only for stochasticity, but also for nonlinear value representations and context-dependent scaling. From a substantive perspective, the ability to represent monetary magnitude appears to be an emergent capability associated with model scale. Smaller models do not reliably encode or compare monetary quantities, limiting their suitability for economic or behavioral tasks involving incentives. Future research should therefore expand the model set to better characterize how representational capacity evolves with scale, explore alternative statistical frameworks robust to deterministic or saturated behavior, and, ultimately, connect behavioral regularities to internal representations through mechanistic analysis of LLM architectures.

References

- Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., & Schulz, E. (2025). Playing repeated games with large language models. *Nature Human Behaviour*, 1–11.
- Anderson, S. P., De Palma, A., & Thisse, J.-F. (1992). *Discrete choice theory of product differentiation*. MIT press.
- Bianchi, F., Chia, P. J., Yuksekgonul, M., Tagliabue, J., Jurafsky, D., & Zou, J. (2024). How well can llms negotiate? negotiationarena platform and analysis. arxiv.
- Bommasani, R., Hudson, D. A., et al. (2022). On the opportunities and risks of foundation models. <https://arxiv.org/abs/2108.07258>
- Bouneffouf, D., & Feraud, R. (2025). Multi-armed bandits meet large language models. *arXiv preprint arXiv:2505.13355*.
- Chen, Y., Liu, T. X., Shan, Y., & Zhong, S. (2023). The emergence of economic rationality of GPT. *Proceedings of the National Academy of Sciences*, 120(51), e2316205120.
- Fan, C., Chen, J., Jin, Y., & He, H. (2024). Can large language models serve as rational players in game theory? a systematic analysis. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16), 17960–17967.
- Fechner, G. T. (1860). *Elemente der psychophysik* (Vol. 2). Breitkopf u. Härtel.
- Fedyk, A., Kakhbod, A., Li, P., & Malmendier, U. (2024). Ai and perception biases in investments: An experimental study. *Available at SSRN 4787249*.
- Gescheider, G. A. (2013). *Psychophysics: The fundamentals*. Psychology Press.
- Guo, F. (2023). GPT in game theory experiments. <https://arxiv.org/abs/2305.05516>
- Horowitz, I., & Plonsky, O. (2025). Llm agents display human biases but exhibit distinct learning patterns. <https://arxiv.org/abs/2503.10248>
- Horton, J. J. (2023). *Large language models as simulated economic agents: What can we learn from homo silicus?* (Tech. rep.). National Bureau of Economic Research.

- Huang, J.-t., Li, E. J., Lam, M. H., Liang, T., Wang, W., Yuan, Y., Jiao, W., Wang, X., Tu, Z., & Lyu, M. (2025). Competing large language models in multi-agent gaming environments. *The Thirteenth International Conference on Learning Representations*.
- Jia, J., Yuan, Z., Pan, J., McNamara, P. E., & Chen, D. (2024). Decision-making behavior evaluation framework for llms under uncertain context. *Advances in neural information processing systems*, 37, 113360–113382.
- Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35, 22199–22213.
- Lai, S., Kim, J., Kunievsky, N., Potter, Y., & Evans, J. (2025). Biased ai improves human decision-making but reduces trust. *arXiv preprint arXiv:2508.09297*.
- Li, L., Yao, Y., Wang, Y., Li, C., Teng, Y., & Wang, Y. (2025). The other mind: How language models exhibit human temporal cognition. <https://arxiv.org/abs/2507.15851>
- Li, N., Gao, C., Li, M., Li, Y., & Liao, Q. (2024). Econagent: Large language model-empowered agents for simulating macroeconomic activities. *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*, 15523–15536.
- Liang, L., & Acuna, D. E. (2020). Artificial mental phenomena: Psychophysics as a framework to detect perception biases in ai models. *Proceedings of the 2020 conference on fairness, accountability, and transparency*, 403–412.
- Manning, B. S., Zhu, K., & Horton, J. J. (2024). *Automated social science: Language models as scientist and subjects* (tech. rep.). National Bureau of Economic Research.
- Olah, C., Satyanarayan, A., Johnson, I., Carter, S., Schubert, L., Ye, K., & Mordvintsev, A. (2018). The building blocks of interpretability. *Distill*, 3(3), e10.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. (2019). Language models are unsupervised multitask learners. *OpenAI blog*, 1(8), 9.
- Ross, J., Kim, Y., & Lo, A. W. (2024). LLM economicus? mapping the behavioral biases of LLMs via utility theory. <https://arxiv.org/abs/2408.02784>
- Schulz, J. F., Fischbacher, U., Thöni, C., & Utikal, V. (2014). Affect and fairness: Dictator games under cognitive load. *Journal of Economic Psychology*, 41, 77–87.
- Wachter, J., Radloff, M., Smolej, M., & Kinder-Kurlanda, K. (2025). Are llms (really) ideological? an irt-based analysis and alignment tool for perceived socio-economic bias in llms. *arXiv preprint arXiv:2503.13149*.
- Weber, E. H. (1834). *De pulsu, resorptione, auditu et tactu: Annotationes anatomicae et physiologicae...* CF Koehler.
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. R., & Cao, Y. (2022). React: Synergizing reasoning and acting in language models. *The eleventh international conference on learning representations*.

Appendix A

Experimental Stimuli and Prompt

Table A1

List of Experimental Stimuli Sample. Payoff matrices across six orders of magnitude ($m \in \{80, \dots, 100000\}$). D scales linearly with m . Only representative trials are shown. d denotes the change in the decision maker’s payoff from choosing the unequal allocation relative to the equal split.

Budget (m)	Option C1 (Equal)		Option C2 (Uneq.)		Gain (d) ($a_2 - a_1$)
	Self	Other	Self	Other	
80	40	40	38.5	41.5	-1.5
	40	40	40.1	39.9	0.1
	40	40	42.5	37.5	2.5
94	47	47	45.5	48.5	-2.5
	47	47	47.1	46.9	0.1
	47	47	49.5	44.5	2.5
100	50	50	48.5	51.5	-1.5
	50	50	50.1	49.9	0.1
	50	50	52.5	47.5	2.5
1,000	500	500	485.0	515.0	-15.0
	500	500	501.0	499.0	1.0
	500	500	525.0	475.0	25.0
10,000	5,000	5,000	4,850	5,150	-150.0
	5,000	5,000	5,010	4,990	10.0
	5,000	5,000	5,250	4,750	250.0
100,000	50,000	50,000	48,500	51,500	-1,500.0
	50,000	50,000	50,100	49,900	100.0
	50,000	50,000	52,500	47,500	2,500.0

Experimental prompt template

All trials are conducted as one-shot decision problems. The model is instructed to ignore any prior conversation history and to make a single, binding choice based solely on the information provided in the current trial. Below we report the exact prompt template used across all experimental conditions, with payoff values instantiated for each trial.

Prompt:

Ignore any previous conversation and memory, and decide based solely on the information below.
 You are in an interactive situation. Your task is to make a final, binding decision. There is one other player, labelled “X”. You and X are both anonymous players. Your decision impacts your earnings and the earnings of X.
 Your task is to make a single decision on how to split a certain amount of money between yourself and X. You have options A and B. They impact payoffs in the following way:
 Option A: You keep {dictatorA}, X gets {recipientA}
 Option B: You keep {dictatorB}, X gets {recipientB}
 Please choose one of the options. Output only “A” or “B”.

Appendix B

Point of Subjective Equality estimates

Table B1

Point of Subjective Equality (PSE) by model and budget. PSE is computed as $-\beta_0/\beta_1$ from logistic regressions of $\Pr(C_1)$ on monetary difference d , estimated separately for each model and budget. p -values smaller than .001 are reported as $< .001$. PSE values become unstable when slope estimates are close to zero and should be interpreted with caution in such cases.

Model	m	n	β_0	β_1	$p(\beta_1)$	PSE
deepseek-r1:70b	80	220	1.78	-0.43	0.0033	4.12
	94	220	3.08	-0.79	$< .001$	3.90
	100	220	1.78	-0.09	0.551	19.97
	1,000	220	2.15	-0.03	0.108	79.64
	10,000	220	2.68	-0.005	0.021	572.02
	100,000	220	1.63	-0.0002	0.082	6670.12
gpt-oss:20b	80	220	0.26	-2.30	$< .001$	0.11
	94	220	0.49	-2.76	$< .001$	0.18
	100	220	0.64	-1.54	$< .001$	0.41
	1,000	219	0.27	-0.23	$< .001$	1.18
	10,000	218	0.39	-0.028	$< .001$	13.83
	100,000	220	0.12	-0.0019	$< .001$	64.92
llama3.1:70b	80	220	0.09	-1.94	$< .001$	0.05
	94	220	0.13	-2.33	$< .001$	0.05
	100	220	-0.32	-5.39	$< .001$	-0.06
	1,000	220	-0.34	-0.33	$< .001$	-1.01
	10,000	220	-0.58	-0.041	$< .001$	-14.04
	100,000	220	-0.07	-0.0023	$< .001$	-28.46
llama3.1:8b	80	220	-0.19	-0.14	0.198	-1.35
	94	220	-0.08	0.19	0.082	0.40
	100	220	-0.03	0.02	0.831	1.30
	1,000	220	-0.22	-0.003	0.764	-68.60
	10,000	220	-0.07	-0.0019	0.088	-40.08
	100,000	220	-0.06	-0.0001	0.370	-642.29